Nathan David Obeng-Amoako, E.I.T.

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RESEARCH INTERESTS

- Transit Signal Priority
- Traffic Simulations / Digital Twins
- Intelligent Transportation Systems
- Multimodal Corridor Planning
- Bicycle Network Analysis
- Geospatial Methods in Transportation

EDUCATION

Doctor of Philosophy in Civil Engineering (Transportation) **Sep 2023 – Present**
Northeastern University, Boston, MA (GPA: 4.00/4.00)
Research Advisor: Professor Peter Furth

Master of Science in Civil Engineering (Transportation) **Sep 2023 – Apr 2025**
Northeastern University, Boston, MA (GPA: 4.00/4.00) **(Conferred Dec 2026)**

Bachelor of Science in Civil Engineering **Aug 2018 – Sep 2022**
Kwame Nkrumah University of Science and Technology (KNUST), Kumasi **(Conferred Nov 2022)**
Recognition: Best Graduating Student, Class of 2022

PROFESSIONAL EXPERIENCE

Transportation Engineering/Planning Intern, Fehr & Peers, Oakland, CA **Apr 2026 – Jun 2026**
- Supported transit signal priority and complete streets projects for state/municipal clients
- Contributed to metro station area studies and transit-oriented development analysis in the East Bay
- Developed federal grant materials for the City of Palo Alto under the OBAG 4 funding cycle
- Conducted GIS and active transportation analysis for corridor and safety planning projects

Program Assistant, Sustainable Transportation Summer Academy, Delft, Holland **Jun 2024 – Aug 2026**
- Provided logistical, academic, and evaluative support to students over three summers
- Assisted in teaching Vision Zero safety principles to 25 Northeastern University students
- Explored car-free and car-lite zone strategies to enhance active/public transportation

Assistant Transportation Engineer, BHM Construction International (UK) Ltd. **Oct 2022 – Jul 2023**
- Investigated traffic problems and proposed recommendations to improve traffic flow and safety
- Conducted 20+ daily quality control tests on construction materials and supported site reporting

Civil Engineering Trainee, China Henan International Corporation Ltd. **Sep 2021 – Dec 2021**
- Supervised 15+ artisans daily to ensure strict adherence to project specifications and timelines
- Contributed to precise structural detailing for reinforced concrete components

SPONSORED RESEARCH PROJECTS

Project: Digital Twin for Adaptive Traffic Signal Control **Aug 2026 – Present**
Funding Agency: StreetIQ, LLC (an AI-powered infrastructure intelligence platform)
- Building and calibrating a Vissim microsimulation of Boston arterial corridors as a testbed for adaptive signal control algorithm development
- Developing external signal control software via Vissim's API to enable iterative, closed-loop algorithm testing and validation against existing control

Project: Bicycle Level of Traffic Stress at Intersections (xLTS) **Jun 2024 – Aug 2025**
Funding Agency: United States Department of Transportation (USDOT)
- Assessing the impact of high-stress crossings on bike network connectivity
- Developing algorithms and data structures for traffic stress network analysis

Project: Accessible Bus Stop Design in the Presence of Bike Lanes **Sep 2023 – Jun 2024**
Funding Agency: Massachusetts Department of Transportation (MassDOT)
- Studied bus rider and bicyclist behavior at floating and constrained bus stops
- Proposed design improvements to reduce bus-rider-and-bicyclist conflicts

PEER-REVIEWED JOURNAL PUBLICATIONS

2025

1. Peter Furth, Ray Saeidi-Razavi, **Nathan David Obeng-Amoako** and Milad Tahmasebi**.** TSP-Friendly Underlying Traffic Signal Control: An Essential Complement of Transit Signal Priority. *Future Transportation* (2025), doi: https://doi.org/10.3390/futuretransp5040160

2. Joshua Kofi Asamoah, Blessing Agyei Kyem, **Nathan David Obeng-Amoako**, Armstrong Aboah, SAAM-ReflectNet: Sign-Aware Attention-Based Multitasking Framework for Integrated Traffic Sign Detection and Retroreflectivity Estimation, *Expert Systems With Applications* (2025), doi: https://doi.org/10.1016/j.eswa.2025.128003

Submitted and under review:

2026

3. Peter G. Furth, **Nathan David Obeng-Amoako***, Armin Akhavan, Bita Sadeghinasr, Ray Saeidi, Luis Miranda-Moreno, Christine Théberge-Barrette, Paul Moser, and Nick Ferenchak. Intersections as Barriers to Low-Stress Cycling: Intersection Level of Traffic Stress and Dual-Graph Network Modeling. *Journal of Transport Geography*.

RESEARCH REPORTS

2024

1. Eleni Christofa, Chengbo Ai, Peter Furth, Yu-Min Yang, Dewan Tanvir Ahammed, and **Nathan David Obeng-Amoako**. (2024). *Accessible Bus Stop Design in the Presence of Bike Lanes*. Massachusetts Department of Transportation (MassDOT), Office of Transportation Planning. Report No. 24-060. Retrieved from https://rosap.ntl.bts.gov/view/dot/79327

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JOURNAL REVIEWS

1. Journal of Transportation Engineering, Part A: Systems, ASCE **2026 – Present**
2. Transportation Research Interdisciplinary Perspectives, Elsevier **2025 – Present**
3. Expert Systems with Applications, Elsevier **2025 – Present**
4. Transportation Research Record, SAGE Publications **2024 – Present**

SELECTED CONFERENCE PRESENTATIONS

2027 1. Furth, P., **Obeng-Amoako, N. D.***, Akhavan, A., Sadeghinasr, B., Saeidi, R., Miranda-Moreno, L., Théberge-Barrette, C., & Moser, P. (2027, January). *Intersections as Barriers to Low-Stress Cycling: Intersection Level of Traffic Stress and Dual-Graph Network Modeling*. Transportation Research Board Annual Meeting, Washington, DC.

2026 2. **Obeng-Amoako, N. D.***, Furth, P., Akhavan, A., Sadeghinasr, B., Saeidi, R., Miranda-Moreno, L., Théberge-Barrette, C., & Moser, P. (2026, August). *Intersections as Barriers to Low-Stress Cycling: Intersection Level of Traffic Stress and Dual-Graph Network Modeling*. Bridging Transportation Researchers (BTR) Conference, Online.

3. Furth, P., & **Obeng-Amoako N. D.*** (2026, May). *Why Johnny and Mary Can't Ride to School: Identifying Barriers to Safe Bicycle Access*. Boston Area Research Initiative Conference, Boston, MA.

4. Yang, Y., Ahammed, D. T., **Obeng-Amoako, N. D.**, Christofa, E., Ai, C., & Furth, P. (2026, January). *Community-Engaged Informed Floating Bus Stop Guidelines* [Poster]. Transportation Research Board Annual Meeting, Washington, DC.

2025 5. **Obeng-Amoako, N. D.**, & Furth, P. (2025, May). *Underlying Traffic Signal Control: The Key to Effective Transit Signal Priority – A Case Study of MBTA Bus Route 39 along South Huntington Avenue, Boston, MA* [Poster]. MassDOT Transportation Innovation Conference, Worcester, MA.

6. Yang, Y., Ahammed, D. T., **Obeng-Amoako, N. D.**, Christofa, E., Ai, C., & Furth, P. (2025, June). *Design Guidelines for Accessible Floating Bus Stops*. International Conference on Transportation and Development (ASCE), Glendale, AZ.

2024 7. Yang, Y., Ahammed, D. T., **Obeng-Amoako, N. D.**, Christofa, E., Ai, C., & Furth, P. (2024, April). *Accessible Bus Stop Design in the Presence of Bike Lanes* [Poster]. MassDOT Transportation Innovation Conference, Worcester, MA.

2023 8. **Obeng-Amoako, N. D.**, Senaya, A. L., Kassim, B., Owusu-Ansah, F., Damoah, E., Worclachie, S., & Tutu, K. A. (2023, April). *Modification of Bitumen with Waste Materials for Enhanced Aggregate Retention in Surface-Dressed Roads*. 3rd IRF Africa Regional Congress & Exhibition, Accra, Ghana.

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APPOINTMENTS AND ADVISORY ROLES

Member, Massachusetts Bicycle and Pedestrian Advisory Board, MassDOT **Jan 2026 – Present**

- Appointed by Governor Maura T. Healey to advise on state-wide bicycle and pedestrian infrastructure policies and programs
- Provide guidance on transportation infrastructure, safety, and accessibility initiatives

NOTABLE DISTINCTIONS AND ACHIEVEMENTS

- Graduate Scholarship Award, Southern New England ITE Section (SENEITE) **Sep 2026**
- Grant Recipient, New England African Business Expo, Worcester, MA **May 2025**
- Civil & Environmental Engineering Distinguished Fellowship Award, Northeastern Univ. **Aug 2023**
- Associated Consultants Ltd. Award for Best Graduating Student, KNUST **Jul 2023**
- Provost Award for Best & Excellent Students in College of Engineering, KNUST (x4) **2019 – 2022**
- Academic Excellence Award, Ghana Engineering Students' Association, Ghana **Sep 2022**
- Global Finalist, Unilever Future Leaders League Business Pitch Competition, U.K. **Jul 2022**
- Scholar, Tullow Oil Scholarship Scheme, Ghana **Apr 2019**
- Semifinalist, National Science and Maths Quiz (NSMQ), Ghana **Jul 2018**

RELEVANT SOFTWARE AND SKILLS

- **Python, R, MATLAB** machine learning, deep learning, advanced data analysis
- **PTV Vissim/Vistro** traffic flow simulations, traffic signal analysis, traffic impact analysis
- **Synchro** traffic flow modeling, traffic signal timing optimization
- **ArcGIS** spatial data mapping and analysis for informed decision-making
- **Gephi** network data analysis and visualization
- **AutoCAD** precision drafting and technical drawings, 2D and 3D
- **Bluebeam Revu** creating, editing, and marking up 2D digital blueprints
- **HTML/CSS/JavaScript** developing web applications
- **Minitab, MS Excel** data visualization, statistical computing, data organization

LEADERSHIP EXPERIENCE

The Ckodon Foundation:

❑ **Executive Director of Partnerships** **May 2026 – Present**

- Lead partnership development with universities, mentors, and sponsor organizations
- Deliver graduate admissions and test-preparation webinars to prospective applicants

❑ **Co-Director for the Graduate Mentorship Program** **Aug 2025 – Apr 2026**

- Co-directed a mentorship program guiding applicants through graduate admissions
- Coordinated mentor recruitment, matching, and program delivery across cohorts

Secretary, Ghana Transportation Professionals Forum (GTPF) **Feb 2026 – Present**

- Maintain membership records and coordinate executive board communications
- Support event logistics and engagements with Ghana transportation stakeholders

Committee Member, Career Mixer Committee, CEE Dep't, Northeastern Univ. **Sep 2025 – Feb 2026**
- Organizing networking events connecting students with industry
- Coordinating logistics for career dev't workshops and employer engagements

Executive C'ttee Member, CEE Graduate Student Council, Northeastern Univ. **Feb 2024 – Present**
- Brainstorming and implementing strategies to enhance graduate student experience
- Serving as a liaison between the departmental chair and the graduate student body

Department Rep., College of Engineering. PhD Council, Northeastern Univ. **Jun 2024 – Aug 2025**
- Meeting with advisor to discuss novel ways of improving student experience
- Organizing weekly coffee hours and semesterly happy hours for PhD students

Communications Director, Young Professionals and Youth Coalition, Ghana **Jan 2021 – Dec 2021**
- Social media management to curate consistent communications content
- Organization of in-person and virtual engagements with members and partners

President, Civil Engineering Students' Association, KNUST **Feb 2020 – Sep 2021**
- Coordination of fundraising events to secure administrative equipment
- Establishment of academic support scheme for underachieving students

TEACHING EXPERIENCE

Northeastern University, Boston, MA

Spring 2026 **Graduate Teaching Assistant, Energy Systems: Science, Technology, and Sustainability**
- Provided grading support for coursework on energy technologies and sustainability
- Gave an invited lecture on Transportation Energy Systems in a Developing World

Spring 2026 **Graduate Teaching Assistant, Transportation Systems Analysis and Planning**
- Supported students during office hours with questions on design software (Synchro)
- Evaluated assignments on intersection analysis, signal timing, and street design principles

Fall 2025 **Graduate Teaching Assistant, Highway Design**
- Provided grading support for homework assignments
- Assisted students during office hours, addressing questions about design principles

Fall 2025 **Graduate Teaching Assistant, Transportation Systems Analysis and Planning**
- Evaluated coursework and provided constructive feedback on analytical assignments
- Supported students' understanding of transportation system planning frameworks

Fall 2023 **Course Instructor, Recitation for Statics and Solid Mechanics**
- Tutored fundamental concepts of Newtonian physics and solid mechanics
- Evaluated students' test performance and offered comprehensive feedback

PROFESSIONAL AFFILIATIONS

- American Society of Civil Engineers (ASCE) **Feb 2025 – Present**
- Boston Society of Civil Engineers Section (BSCES) of ASCE **Feb 2025 – Present**
- New England Intelligent Transportation Society (NEITS) **Sep 2023 – Present**
- Institute of Transportation Engineers (ITE) **Jan 2024 – Present**

VOLUNTEER EXPERIENCE & COMMUNITY INVOLVEMENT

- **Judge**, Massachusetts Science and Engineering Fair, Worcester (April 2025)
- **Cycling Activist**, Boston Cycling Union, Boston, MA (Feb 2025 – Present)
- **Executive Vice President and Co-Founder**, Student Energy Society, KNUST (Jan 2022 – Oct 2022)
- **Community Outreach Coordinator**, Nat'l Society of Black Engineers, KNUST (Jan 2022 – Sep 2022)
- **Project Coordinator**, Engineers Without Borders, KNUST (Aug 2021 - Jul 2022)

BLOG POSTS

1. "Ghana is Moving, but Barely: The Case for a Smarter Public Transportation Strategy." MyJoyOnline, April 2026. https://www.myjoyonline.com/ghana-is-moving-but-barely-the-case-for-a-smarter-public-transportation-strategy/

2. "Replacing the Reid Overpass with an At-Grade Intersection (Cambridge, near BU Bridge)." Co-authored with Peter G. Furth. January 2026. https://peterfurth.sites.northeastern.edu/2026/01/06/replacing-the-reid-overpass-with-an-at-grade-intersection-cambridge-near-bu-bridge/

3. "PhD Student Reflects on the Sustainable Transportation Dialogue of Civilizations in Holland." Northeastern University College of Engineering News, August 2024. https://coe.northeastern.edu/news/a-reflection-on-the-sustainable-transportation-dialogue-of-civilizations-course-in-holland/

4. "Bus-Friendly Traffic Signals Can Reduce Bus Delay By 90% on South Huntington Avenue." Co-authored with Peter G. Furth. April 2024. https://peterfurth.sites.northeastern.edu/2024/06/04/bus-friendly-traffic-signals-can-reduce-bus-delay-by-90-on-south-huntington-avenue/

REFERENCES

Available upon request